KAVIBARATHI M

B.Tech Artificial Intelligence & Data Science | AI/ML | Python | Data Structures & Algorithms
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PROFESSIONAL SUMMARY

Pre-final year B.Tech Artificial Intelligence & Data Science student with a strong foundation in Python, Data Structures & Algorithms, Machine Learning, and Data Analytics. Experienced in building AI-driven solutions for queue management, image recognition, and legacy system modernization using NumPy, Pandas, Firebase, FastAPI, PostgreSQL, and Streamlit. Proven ability to translate real-world problems into working AI models through hackathons and academic projects.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, Data Structures & Algorithms
Data Science: NumPy, Pandas, Exploratory Data Analysis (EDA), Data Analytics, Data Visualization, Matplotlib
AI/ML: Machine Learning, Deep Learning, Neural Networks, Generative AI, Retrieval-Augmented Generation (RAG)
Tools & Platforms: Streamlit, FastAPI, Firebase, PostgreSQL, Git, GitHub, VS Code

EDUCATION

B.Tech, Artificial Intelligence & Data Science
Dhanalakshmi Srinivasan College of Engineering and Technology | CGPA: 8.19 | Expected Graduation: 2028

KEY PROJECTS

AI Skill Gap Detector
- Built an AI tool that compares a user's current skill set against target job roles to identify gaps and recommend a personalized upskilling path.
- Used Python-based data analysis to score skill matches and generate actionable, role-specific career recommendations.

AI-Based Smart Queue Prediction & Management System
- Developed a machine learning model that predicts customer queue waiting times from historical data to support smarter staff and resource allocation.
- Implemented the data pipeline with Python, NumPy, and Pandas, improving predictability of service times for queue-heavy environments (e.g., banks, hospitals).

Image-Based Cattle Breed Recognition — *Team BOVINETECH, Smart India Hackathon (Internal)*
- Built an AI image-recognition system to automatically identify cattle breeds from photographs, supporting livestock management and breeding decisions.
- Collaborated in a 4-member team under hackathon time constraints, applying computer vision and deep learning concepts to an agricultural use case.

Legacy Application Modernization Using AI — *IBM SkillsBuild Hackathon*
- Contributed to an AI-assisted approach for modernizing legacy software, helping identify outdated components and propose upgrade strategies.
- Gained hands-on exposure to applying AI within enterprise software modernization workflows as part of a hackathon team.

INTERNSHIP EXPERIENCE

AI for Sustainability Virtual Internship — *1M1B x AICTE x IBM SkillsBuild* *Dec 2025 – Jan 2026*
- Completed a virtual internship applying AI techniques to sustainability-focused real-world problems.

Internship — *National Small Industries Corporation (NSIC)* *Jun 2026 (1 week)*
- Completed technical training with exposure to professional working environments and project workflows.

CERTIFICATIONS & ACHIEVEMENTS

- First Prize — Idea Pitching for “AI-Powered Autonomous Multi-Terrain Waste Collection and Segregation Robot”
- Second Prize — National-Level GAMEATHON Competition for FitnessStride App
- NPTEL — AI Industrial Management: 70% (Elite)

LEADERSHIP & RESPONSIBILITIES

- Joint Secretary, AI & Data Science Department Association — coordinated departmental academic and technical activities.

ADDITIONAL INFORMATION

Languages: Tamil, English